

Case Study: Customer Loan Default Prediction Using Feature Engineering and Feature Selection

Total Marks: 40

Problem Statement

A financial institution wants to build a machine learning model to predict whether a customer will **default on a loan**. The organization has collected customer demographic, financial, behavioral, and transaction-related data.

However, the dataset contains:

- Missing values in numeric and categorical columns
- Outliers in income and loan amount
- Class imbalance because only a small percentage of customers default
- Highly correlated features
- Irrelevant or low-variance features
- Time-based records that require careful validation
- Customer group information that may cause data leakage

Your task is to perform a complete **feature engineering and feature selection pipeline** before training and evaluating machine learning models.

Dataset Description

Assume the dataset is named:
loan_default_dataset.csv

Dataset Size

Rows: 10,000 customers
Columns: 28 features + 1 target column

Target Variable

Column Name	Description
default_status	Target variable. 1 = Customer defaulted, 0 = Customer did not default

Feature Columns

Column Name	Type	Description
customer_id	ID	Unique customer identifier
application_date	Date	Date of loan application
branch_id	Categorical	Bank branch where loan was issued
customer_group_id	Categorical	Group/family/business identifier
age	Numerical	Age of applicant
gender	Categorical	Male, Female, Other
marital_status	Categorical	Single, Married, Divorced

education_level	Categorical	School, Graduate, Postgraduate
employment_type	Categorical	Salaried, Self-employed, Unemployed
annual_income	Numerical	Annual income of applicant
monthly_expenses	Numerical	Average monthly expenses
loan_amount	Numerical	Loan amount requested
loan_tenure_months	Numerical	Loan repayment period
interest_rate	Numerical	Loan interest rate
credit_score	Numerical	Credit score of applicant
existing_loans_count	Numerical	Number of existing loans
credit_card_utilization	Numerical	Credit card usage percentage
late_payment_count	Numerical	Number of previous late payments
bank_balance	Numerical	Current bank balance
savings_account_balance	Numerical	Savings account balance
transaction_frequency	Numerical	Monthly transaction count
avg_transaction_value	Numerical	Average transaction amount
last_payment_delay_days	Numerical	Delay in most recent payment
property_ownership	Categorical	Own, Rent, Family Owned
city_tier	Categorical	Tier 1, Tier 2, Tier 3
has_insurance	Categorical	Yes, No
loan_purpose	Categorical	Education, Home, Business, Personal
risk_segment	Categorical	Low, Medium, High
default_status	Binary	Target column

Case Study Tasks and Marks Distribution

Section A: Data Understanding and Preprocessing

Marks: 4

Task A1: Load and inspect the dataset

Marks: 2

You should identify:

- Number of rows and columns
 - Numerical and categorical variables
 - Missing value percentage
 - Target class distribution
 - Duplicate records
-

Task A2: Identify feature types

Marks: 2

Classify the columns into:

- Numerical features
- Categorical features
- Date/time features
- ID/group columns
- Target column

Expected output format:

Numerical Features:

age, annual_income, loan_amount, credit_score, ...

Categorical Features:

gender, employment_type, loan_purpose, ...

Date Feature:

application_date

Group Feature:

customer_group_id

Target:

default_status

Section B: Missing Value Treatment**Marks: 8**

The dataset contains missing values in both numerical and categorical columns.

Example missing values:

Column	Missing %
annual_income	8%
credit_score	11%
employment_type	5%
loan_amount	4%
bank_balance	7%
risk_segment	3%

Task B1: Simple Imputation**Marks: 3**

Apply the following simple imputation techniques:

1. Mean Imputation

Recommended columns:

age

loan_tenure_months

interest_rate

2. Median Imputation

Use median imputation for skewed numerical columns.

Example:

Recommended columns:

annual_income

loan_amount

bank_balance

monthly_expenses

3. Mode Imputation

Use mode imputation for categorical columns.

Example:

Recommended columns:

employment_type

gender

marital_status

loan_purpose

risk_segment

Task B2: Advanced Imputation Techniques

Marks: 5

Apply and compare the following techniques.

4. KNN Imputation

Use KNN Imputer for numerical columns.

Explain:

- Why scaling may be needed before KNN imputation
- How nearest neighbors are used to estimate missing values

5. Regression Imputation

Use regression to predict missing credit_score.

Explain:

- Which feature was imputed
- Which predictors were used
- Limitation of regression imputation

6. Iterative Imputer / MICE

Use Iterative Imputer for multiple missing numerical features.

Explain:

- How each missing feature is modeled using other features
 - Why MICE is useful when many columns have missing values
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7. Forward Fill and Backward Fill

Use these methods for time-based data sorted by application_date.

Explain:

- Why forward/backward fill is suitable for time-ordered data
 - Why it should not be used randomly on unordered data
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Section C: Data Transformation and Scaling

Marks: 4

Task C1: Apply Z-score Standardization

Marks: 4

Explain why scaling is important for:

- KNN
 - Logistic Regression
 - Ridge Regression
 - Lasso Regression
 - SVM
 - Distance-based algorithms
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Section D: Handling Imbalanced Data

Marks: 5

The target variable is imbalanced.

Example:

Non-default customers: 92%

Default customers: 8%

Task D1: Analyze class imbalance

Marks: 1

Explain why accuracy is not a reliable metric for this problem.

Task D2: Apply Random Oversampling

Marks: 1

Task D3: Apply Random Undersampling

Explain the main risk of undersampling.

Task D4: Apply SMOTE

Marks: 2

Explain:

- How SMOTE creates synthetic samples
 - Why SMOTE may be better than simple oversampling
 - One limitation of SMOTE
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Section E: Regularization**Marks: 4****Task E1: Apply Ridge Regression / Ridge Classifier****Marks: 2**

Explain:

- Meaning of L2 penalty
- Effect of increasing alpha
- Why Ridge reduces overfitting

Task E2: Apply Lasso Logistic Regression**Marks: 2**

Explain:

- Meaning of L1 penalty
- Why Lasso can perform feature selection
- Difference between Ridge and Lasso

Section F: Cross-Validation Techniques**Marks: 6**

Apply the following validation strategies and explain when each is suitable.

Task F1: K-Fold Cross-Validation**Marks: 1****Task F2: Stratified K-Fold****Marks: 1****Task F3: LOOCV****Marks: 0.5****Task F4: Leave-P-Out****Marks: 0.5****Task F5: Shuffle Split****Marks: 1****Task F6: Time Series Split****Marks: 1****Task F7: Group K-Fold****Marks: 0.5****Task F8: Repeated K-Fold****Marks: 0.5****Section G: Feature Selection****Marks: 7****Task G1: Filter Methods****Marks: 2**

Apply at least two filter methods.

1. Variance Threshold**2. Correlation-Based Selection****3. Chi-Square Test****4. Mutual Information**

Task G2: Wrapper Methods**Marks: 2**

Apply at least two wrapper methods.

1. Forward Selection

2. Backward Elimination

3. Recursive Feature Elimination

4. Exhaustive Feature Selection

Explain why exhaustive feature selection is computationally expensive.

Expected explanation:

Task G3: Embedded Methods**Marks: 2**

Apply at least two embedded feature selection techniques.

1. Elastic Net

2. Decision Tree

3. Random Forest

4. XGBoost

Explain:

- How tree-based models calculate feature importance
- Why embedded methods are faster than wrapper methods

Task G4: Metaheuristic-Based Feature Selection**Marks: 1**

Explain how the following methods can be used for feature selection.

Genetic Algorithm

Expected explanation:

Simulated Annealing

Expected explanation:

Section H: Final Model Building and Evaluation**Marks: 2**

Train at least two models using the selected features.

Logistic Regression

Ridge Classifier

Random Forest

XGBoost

Confusion Matrix

rather than only accuracy.

Final Deliverables

Students must submit:

1. Cleaned dataset after imputation
2. Python notebook or script
3. Feature engineering summary
4. Cross-validation comparison table
5. Imbalance handling comparison
6. Feature selection comparison
7. Final model performance report
8. Conclusion with best technique combination

Marking Scheme Summary

Section	Topic	Marks
A	Data understanding and preprocessing	4
B	Missing value treatment	8
C	Z-score scaling	4
D	Handling imbalanced data	5
E	Regularization	4
F	Cross-validation techniques	6
G	Feature selection	7
H	Final model evaluation	2
	Total	40